

Valerie Zhao

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EDUCATION

University of Washington, Seattle

Sep.2024 - Jun.2026(Expected)

Master of Science in Information Management (Specialization: Data Science & Artificial Intelligence) GPA: 3.9/4.0

China Pharmaceutical University, Nanjing

Sep.2020 - Jun.2024

Bachelor of Science in Information Management and Information Systems

GPA: 3.7/4.0

SKILLS

Programming Languages: Java, Python, JavaScript, SQL, C++, TypeScript, HTML/CSS, Bash

Frameworks & Tools: Spring Boot, React, Vue, Hadoop, Apache Spark, TensorFlow, PyTorch, Git, AuthN/AuthZ

Databases & Caching: MySQL, SQL Server, Redis

Cloud & DevOps: AWS, Azure, Docker, Linux, CI/CD, Nginx

Concepts: Distributed Systems, Microservices, System Design, Data Structures, Software Development Life Cycle

WORK EXPERIENCE

IpserLab - Full-Stack Intern

Jun.2025 - Present

- Engineered modular backend services for a **SaaS** platform using **Java**, **JavaScript**, and **MySQL**, with a custom ORM layer for persistence and implemented **RESTful APIs** to enable reliable JSON-based data exchange.
- Built responsive user/admin interfaces with **React.js** and **TailwindCSS**, including dynamic multi-step forms and state management, and integrated **Chart.js** for interactive data visualization.
- Implemented **secure authentication system** with email validation, server-side login, token-based session management, and 2FA support.
- Designed and launched a 6-step dynamic application workflow supporting applicant details, project data, multi-file uploads (PDF, PNG, JPG, DOC), and multi-currency payment integration (USD & Zimbabwean Dollar).
- Delivered usability features including draft-saving & progress tracking, and integrated Email/SMS notifications for automated status updates.

Legend Biotech - IT Department - Full-Stack Intern

Feb.2023 - Apr.2023

- Developed a cross-platform mobile attendance check-in application using **Java** and **React**, improving employee tracking efficiency.
 - Contributed to a tumor treatment tracking system using **Spring Boot** for managing patient data, treatment schedules, and progress tracking, and collaborating on frontend development using **HTML5**, **CSS3**, and **JavaScript**.
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PROGRAM EXPERIENCE

Seattle Source Ranker

Sep.2025 - Present

- Built a scalable distributed ingestion pipeline with **Celery + Redis** to collect and validate **432K+ repositories from 28K Seattle developers**, achieving **8.1× throughput** and **99.98% reliable ingestion** under strict rate limits.
- Designed a three-phase hybrid API workflow (**GraphQL → REST → GraphQL+HEAD**) to correct GitHub's inconsistent watchers data and remove ~2% invalid repos, delivering high-precision metadata for all ranked projects.
- Implemented a fault-tolerant token rotation system (6 rotating tokens × 5000 req/hr) with intelligent backoff and idempotent retries, reducing rate-limit failures by **92%** and enabling **60–90 minute** full dataset rebuilds.
- Developed a multi-factor ranking engine using log-scaled popularity metrics, project age, issue health, and activity recency; supports **10 languages** plus Python-specific **PyPI tier bonuses (+15.5%)**.
- Integrated PyPI's **700K+** package index and global Top-15K rankings, identifying 1,075 Seattle PyPI packages and highlighting 28 globally significant projects via offline matching with 100% precision.
- Automated daily dataset updates and static deployment via GitHub Actions, generating **9,632 optimized JSON datasets** consumed by a React interface with real-time search, suggestions, lazy pagination, and **<50ms load times**.

Virtual Character Letter Platform

Nov.2024 - Apr.2024

- Developed a hybrid backend using **Spring Boot** for user management and role creation, leveraging **RESTful APIs** for front-end and back-end communication and integrating **LangChain** for LLM/NLP support.
- Utilized **Redis** for caching to reduce latency and designed a multi-database architecture, using **MongoDB** for storing unstructured letter content and **MySQL** for managing structured user and role data.
- Implemented **RabbitMQ** to handle asynchronous messaging between Java and Python services, ensuring efficient communication across the system.
- Deployed the application to an **Amazon EC2 instance**, using **Docker** for containerization and **Nginx** for reverse proxy management. Configured **AWS CloudWatch** for monitoring and logging to ensure system stability.